US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication	20250278038
Kind Code	A1
Publication Date	September 04, 2025
Inventor(s)	NINOMIYA; Yosuke

DEVELOPER CONTAINER AND IMAGE FORMING APPARATUS

Abstract

A developer container includes: an accommodation portion that accommodates a developer; a discharge port through which the developer is discharged from an outer peripheral portion of the accommodation portion on a front end side in a mounting direction; an opening and closing portion that opens and closes the discharge port from an outside; an attachment portion that surrounds the discharge port and to which the opening and closing portion is movably attached, in which the attachment portion has an extending portion extending toward the front end side in the mounting direction, and the extending portion is provided with a support portion that supports the extending portion on a surface opposite to the opening and closing portion.

Inventors: **NINOMIYA; Yosuke (Kanagawa, JP)**

Applicant: **FUJIFILM Business Innovation Corp.** (Tokyo, JP)

Family ID: **92583023**

Assignee: **FUJIFILM Business Innovation Corp. (Tokyo, JP)**

Appl. No.: **18/822294**

Filed: **September 02, 2024**

Foreign Application Priority Data

JP	2024-030053	Feb. 29, 2024

Publication Classification

Int. Cl.: **G03G15/08** (20060101); **G03G15/01** (20060101)

U.S. Cl.:

CPC **G03G15/0872** (20130101); **G03G15/0121** (20130101); **G03G15/0189** (20130101);

G03G15/0886 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
[0001] This application is based on and claims priority under 35 USC 119 from Japanese Patent Application No. 2024-030053 filed Feb. 29, 2024.
BACKGROUND
(i) Technical Field
[0002] The present invention relates to a developer container and an image forming apparatus.
(ii) Related Art
[0003] JP2020-170150A (Paragraph 0053 and FIG. 4) discloses a powder container including a toner bottle that accommodates a toner, an outlet port through which the toner flows out, a shutter that opens and closes the outlet port from an outside, and an attachment portion to which the shutter is movably attached.
SUMMARY
[0004] There is a developer container including an attachment portion to which an opening and closing portion that opens and closes a discharge port for a developer is movably attached on a front end side in a mounting direction of an accommodation portion for the developer. In such a developer container, from the viewpoint of securing sealability of the discharge port by the opening and closing portion at the attachment portion, widening a flat surface portion of a portion of the attachment portion facing the opening and closing portion is desired.
[0005] Aspects of non-limiting embodiments of the present disclosure relate to a developer container that enables, regarding an extending portion of the attachment portion that extends toward a front end side in a mounting direction, compared to a case where the extending portion is not provided with a support portion that supports the extending portion, the extending portion to become a portion that is less likely to be deformed or damaged by providing a flat surface portion that is made wider to secure the sealability of the discharge port by the opening and closing portion.
[0006] Aspects of certain non-limiting embodiments of the present disclosure address the above advantages and/or other advantages not described above. However, aspects of the non-limiting embodiments are not required to address the advantages described above, and aspects of the non-limiting embodiments of the present disclosure may not address advantages described above.
[0007] According to an aspect of the present disclosure, there is provided a developer container including: [0008] an accommodation portion that accommodates a developer; [0009] a discharge port through which the developer is discharged from an outer peripheral portion of the accommodation portion on a front end side in a mounting direction; [0010] an opening and closing portion that opens and closes the discharge port from an outside; an attachment portion that surrounds the discharge port and to which the opening and closing portion is movably attached, [0011] in which the attachment portion has an extending portion extending toward the front end side in the mounting direction, and [0012] the extending portion is provided with a support portion that supports the extending portion on a surface opposite to the opening and closing portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0013] Exemplary embodiment(s) of the present invention will be described in detail based on the following figures, wherein:
[0014] FIG. **1** is a schematic view of an image forming apparatus including a developer container

according to a first exemplary embodiment;
[0015] FIG. **2** is a schematic view showing a state at the time of mounting of a developer container and a mounting portion;
[0016] FIG. **3** is a perspective view of the developer container according to the first exemplary embodiment;
[0017] FIG. **4** is an exploded perspective view of the developer container in FIG. **3**;
[0018] FIG. **5** is a perspective view showing a portion of the developer container on a front end side in a mounting direction;
[0019] FIG. **6** is a perspective view including a cross section taken along line VI-VI of the developer container in FIG. **5**;
[0020] FIG. **7**A is a bottom perspective view showing a state where an opening and closing portion of a portion of the developer container on the front end side in the mounting direction is closed, and FIG. **7**B is a bottom perspective view obliquely showing a state where the opening and closing portion is opened at the portion in FIG. **7**A;
[0021] FIG. **8** is a perspective view showing a part of the developer container and the mounting portion;
[0022] FIG. **9** is a perspective view of an opening and closing guide body in the mounting portion;
[0023] FIG. **10** is a perspective view of an end portion holding body of the developer container;
[0024] FIG. **11**A is a front view showing a state where the developer container is mounted in the opening and closing guide body in the mounting portion, and FIG. **11**B is a front view including a partial cross section showing a state where a part of the opening and closing guide body is omitted in the state of FIG. **11**A;
[0025] FIG. **12** is a side view including a cross section taken along line VI-VI of the developer container in FIG. **5**; and
[0026] FIG. **13** is a front view showing another configuration example of a support portion and an identification shape portion.
DETAILED DESCRIPTION
[0027] Hereinafter, exemplary embodiments for carrying out the present invention will be described.
First Exemplary Embodiment
[0028] FIG. **1** is a schematic view of an image forming apparatus **1** including a developer container **5** according to a first exemplary embodiment of the present invention.
[0029] In the present specification and the drawings, substantially the same constituent elements are denoted by the same reference numerals. In addition, in the present specification, the duplicated description of the same constituent element will be omitted.
(1) Image Forming Apparatus
[0030] As shown in FIG. **1**, in the image forming apparatus **1**, an image forming section **20** that forms an image consisting of a developer on paper **9** and a paper supply section **40** that supplies the paper **9** to the image forming section **20** are disposed inside a housing **10**.
[0031] In addition, the image forming apparatus **1** is provided with a discharge and accommodation portion **12** that discharges and accommodates the paper **9** on which the image is formed, for example, in an upper portion of the housing **10**.
[0032] Furthermore, the image forming apparatus **1** is provided with a mounting portion **15**, in which the developer container **5** is attachably and detachably mounted, on a part of the housing **10**.
[0033] The image forming section **20** has a plurality of image forming units **21**A, **21**B, **21**C, and **21**D, an intermediate transfer unit **30**, and a fixing unit **36**.
[0034] In the image forming units **21**A, **21**B, **21**C, and **21**D, images of predetermined colors (A, B, C, and D) consisting of the developer are respectively formed on photosensitive drums **22** by an electrophotographic method.
[0035] Examples of the predetermined colors (A, B, C, and D) include colors such as black, cyan,

magenta, and yellow.
[0036] Specifically, in each of the image forming units **21**A, **21**B, **21**C, and **21**D, the corresponding photosensitive drum **22** rotates in a direction indicated by an arrow.
[0037] Then, an outer peripheral surface of each of the photosensitive drums **22** is charged to a predetermined surface potential by a corresponding charging devices **23**.
[0038] Subsequently, on the charged outer peripheral surface of each of the photosensitive drums **22**, image exposure of each color component corresponding to image information is performed by each of latent image forming devices **24**. Accordingly, electrostatic latent images of the predetermined color components are distributed and exclusively formed on the outer peripheral surfaces of the respective photosensitive drums **22**.
[0039] The image information is input from an external source such as an external connection device (not shown) connected to the image forming apparatus **1**.
[0040] Subsequently, in the photosensitive drums **22**, the electrostatic latent images formed on the outer peripheral surfaces thereof are respectively developed by the developers of the corresponding colors respectively supplied from developing devices **25**A, **25**B, **25**C, and **25**D.
[0041] As the developer, for example, a two-component developer containing a toner and a carrier is used. In a case where the developer is the two-component developer, a toner image consisting of a toner of a predetermined color (A, B, C, or D) is obtained as a developed image.
[0042] Since the developing devices **25** differ in that the developer containing the toner of the predetermined color (A, B, C, or D) is used separately, the developing devices **25** are displayed as developing devices **25**A, **25**B, **25**C, and **25**D.
[0043] Accordingly, the toner images of the predetermined colors are respectively formed on the outer peripheral surfaces of the corresponding photosensitive drums **22**.
[0044] In the intermediate transfer unit **30**, the toner images of the predetermined colors respectively formed by the image forming units **21**A, **21**B, **21**C, and **21**D are temporarily held on an intermediate transfer belt **31** by primary transfer, followed by secondary transfer to the paper **9**.
[0045] Specifically, the intermediate transfer belt **31** is rotatably stretched around a plurality of rolls and is rotated in a direction indicated by an arrow.
[0046] Subsequently, the toner images on the photosensitive drums **22** are sequentially primarily transferred to an outer peripheral surface of the intermediate transfer belt **31** by a primary transfer device **33**.
[0047] Thereafter, the intermediate transfer belt **31** transports the toner image primarily transferred to the outer peripheral surface thereof to a secondary transfer position facing a secondary transfer device **34**. At the secondary transfer position, the toner image on the intermediate transfer belt **31** is secondarily transferred to the paper **9** supplied from the paper supply section **40** by the secondary transfer device **34**.
[0048] In the paper supply section **40**, the paper **9** is sent one by one from an accommodation body **41** in which the paper **9** is accommodated to the image forming section **20** by a sending device **42**.
[0049] There are cases where a plurality of sets of the accommodation body **41** and the sending device **42** are provided.
[0050] The paper **9** sent by the sending device **42** is transported through a paper feeding path portion of a transport path Tr constituted by a predetermined number of transport roll pairs **46***a* and **46***b*, paper guide members (not shown), and the like.
[0051] At this time, the paper **9** is sent to the secondary transfer position of the intermediate transfer unit **30** by the transport roll pair **46***b* in synchronization with the secondary transfer, and the toner image is secondarily transferred to one surface of the paper **9**. The paper **9** after the secondary transfer of the toner image is sent from the secondary transfer position toward a fixing unit **36**
[0052] In the fixing unit **36**, the paper **9** on which the toner image has been secondarily transferred is introduced into a fixing processing portion (nip portion) formed between a heating rotating body

37 and a pressurizing rotating body **38**, which rotate in a pressure contact state inside the housing.
[0053] Accordingly, an unfixed toner image is heated and pressurized when passing through the fixing processing portion, and is fixed on the paper **9** as an image.
[0054] As described above, the image consisting of the developer (toner) is formed on one surface of the paper **9**.
[0055] Finally, in the image forming apparatus **1**, the paper **9** on which the image is formed is transported and accommodated so that the paper **9** is discharged to the discharge and accommodation portion **12** through a discharge path portion of the transport path Tr.
[0056] The discharge path portion of the transport path Tr is configured with a predetermined number of transport roll pairs **46***c* and **46***d*, paper guide members (not shown), and the like.
Mounting Portion of Developer Container
[0057] In the image forming apparatus **1**, as the developer container **5**, as shown in FIG. **1**, four developer containers **5**A, **5**B, **5**C, and **5**D in which the developers of the predetermined colors (A, B, C, and D) are exclusively accommodated are used.
[0058] The image forming apparatus **1** is provided with mounting portions **15**A, **15**B, **15**C, and **15**D for attachably and detachably mounting the developer containers **5**A, **5**B, **5**C, and **5**D, respectively, in a part of the housing **10**.
[0059] As shown in FIG. **1**, the mounting portions **15**A, **15**B, **15**C, and **15**D in the first exemplary embodiment are provided in a portion between the intermediate transfer unit **30** and the discharge and accommodation portion **12**.
[0060] In addition, as shown in FIG. **2**, the mounting portions **15**A, **15**B, **15**C, and **15**D are formed as storage space sections that are depressed from a front surface, which is one of side surfaces of the housing **10**, toward a rear side (in a direction indicated by an arrow Z).
[0061] Furthermore, as shown in FIG. **2**, the mounting portion **15**A or the like is in a state of being opened to the outside and in a state of being closed from the outside by an opening and closing door **13** that is rotatably moved with a lower end as a fulcrum on a front surface side of the housing **10**.
[0062] The developer containers **5**A, **5**B, **5**C, and **5**D are attachable to and detachable from the mounting portions **15**A, **15**B, **15**C, and **15**D by being operated to be inserted into and removed from the mounting portions **15**A, **15**B, **15**C, and **15**D in directions indicated by arrows D**1** and D**2**. The direction indicated by the arrow D**1** in FIG. **2** and the like indicates a mounting direction (insertion direction), and the direction indicated by the arrow D**2** indicates a detachment direction (pull-out direction).
[0063] Details of the developer containers **5**A, **5**B, **5**C, and **5**D will be described later.
[0064] In addition, in the image forming apparatus **1**, the developer is consumed and reduced in amount in the developing devices **25**A, **25**B, **25**C, and **25**D by an image forming operation.
[0065] Therefore, in a case where the developer is reduced to a predetermined amount, the developing devices **25**A, **25**B, **25**C, and **25**D are respectively replenished with the developers (toner in this example) of the corresponding colors (A, B, C, and D) from the developer containers **5**A, **5**B, **5**C, and **5**D.
[0066] In the mounting portions **15**A, **15**B, **15**C, and **15**D, a mounting platform **16** and a replenishing device **17** are each disposed inside storage spaces in which the developer containers **5**A, **5**B, **5**C, and **5**D are respectively stored, as shown in FIG. **1**.
[0067] The mounting platform **16** is a structure in which the developer containers **5**A, **5**B, **5**C, and **5**D are mounted and held in the storage space, respectively. The replenishing device **17** is a device that sends out and replenishes a predetermined amount of the developer discharged from the developer containers **5**A, **5**B, **5**C, and **5**D to the corresponding developing devices **25**A, **25**B, **25**C, and **25**D.
[0068] In addition, a replenishing transport path **18** is provided in each of the mounting portions **15**A, **15**B, **15**C, and **15**D.

[0069] The replenishing transport path **18** is a developer transport path disposed to connect the replenishing devices **17** to the corresponding developing devices **25**A, **25**B, **25**C, and **25**D.

(2) Developer Container

[0070] As shown in FIG. **3**, each of the developer containers **5**A, **5**B, **5**C, and **5**D includes an accommodation portion **51** that accommodates a predetermined developer G (see FIG. **6** and the like).

[0071] In a case where the developer containers **5**A, **5**B, **5**C, and **5**D accommodate only toner T (see FIG. **6** and the like) of the two-component developer as the developer G, the developer container **5**A, **5**B, **5**C, and **5**D are also referred to as, for example, a toner cartridge.

[0072] As shown in FIG. **4**, the accommodation portion **51** in the first exemplary embodiment is constituted by an accommodation main body **52**, an end portion holding body **53**, and an agitating body **54**.

[0073] The accommodation portion **51** has a structure in which the agitating body **54** is attached to one end portion of the accommodation main body **52** and the portion to which the agitating body **54** is attached is rotatably held by the end portion holding body **53**.

Accommodation Main Body

[0074] The accommodation main body **52** is a portion that accommodates the developer G, at least the toner T in this example.

[0075] The accommodation main body **52** in the first exemplary embodiment has a substantially cylindrical structure as a whole in which one end is open and the other end is closed. That is, in the accommodation main body **52**, an end portion **52***a* on a downstream side in the mounting direction D**1** of the accommodation portion **51**, and thus the developer container **5**, is open, and an end portion **52***b* on an upstream side in the mounting direction D**1** is closed.

[0076] In addition, the accommodation main body **52** is configured to rotate in a predetermined direction M in a case where the accommodated developer G (toner T) is discharged during replenishment.

[0077] Incidentally, the mounting direction D**1** of the developer containers **5**A, **5**B, **5**C, and **5**D is a direction along a longitudinal direction of the accommodation main body **52** having a form elongated in one direction.

[0078] In addition, the accommodation main body **52** is provided with a coupled portion **523** to which the agitating body **54** is coupled, at the end portion **52***a* that is open. The coupled portion **523** is a structural portion with, for example, threads formed on an outer peripheral surface of the accommodation main body **52**.

[0079] In addition, the accommodation main body **52** is provided with a grip portion **527** having a shape protruding from the end portion **52***b* toward the upstream side in the mounting direction D**1**, at the end portion **52***b*. The grip portion **527** is used as a portion that is gripped and held by a user by hand in a case where the developer container **5** is mounted in the mounting portion **15**.

[0080] Furthermore, the accommodation main body **52** is provided with a groove **524** that extends in a helical shape on the outer peripheral surface thereof. As shown in FIG. **6** and the like, the groove **524** is formed with a ridge portion **525** on an inner wall surface of the accommodation main body **52**, the ridge portion **525** extending in a helical shape to be raised inwardly and to substantially correspond to the shape of the groove **524**.

[0081] The helical ridge portion **525** functions as transporting means that transports the accommodated developer G (toner T) to be moved toward a discharge port **55**, which will be described below, when the accommodation main body **52** is rotated in the predetermined direction M.

End Portion Holding Body

[0082] The end portion holding body **53** is a portion that rotatably holds the end portion **52***a* of the accommodation main body **52** via the agitating body **54**.

[0083] In addition, the end portion holding body **53** also serves as a portion on a front end side in

the mounting direction D**1** of the developer container **5** or the accommodation portion **51**.
[0084] As shown in FIGS. **5** and **6**, the end portion holding body **53** in the first exemplary embodiment has a cylindrical main body portion **531**.
[0085] An internal space of the main body portion **531** is divided into front and rear sections by a partition wall **532** in the mounting direction D**1**.
[0086] As shown in FIG. **6**, the main body portion **531** has a first internal space S**1** that is present on the upstream side in the mounting direction D**1** from the partition wall **532**.
[0087] The first internal space S**1** is a space that accommodates the developer G (toner T) while accommodating a part of the accommodation main body **52** to be held via the agitating body **54**.
[0088] The partition wall **532** is an end portion on the downstream side in the mounting direction D**1** in an accommodation space of the accommodation portion **51** in which the developer G (toner T) is accommodated. In other words, the partition wall **532** also serves as a portion that functions as a lid that closes the open end portion **52***a* of the accommodation main body **52** together with the first internal space S**1** in a state where the agitating body **54** is interposed.
[0089] In addition, the main body portion **531** has a second internal space S**2** that is present on the downstream side in the mounting direction D**1** from the partition wall **532**.
[0090] The second internal space S**2** is a space in which a shaft coupling **58** and the like on a driven side are disposed.
[0091] The shaft coupling **58** on the driven side is coupled to a shaft coupling **19** (see FIG. **8**) on a drive side, which is detachably connected to transmit rotational power disposed on a mounting portion **15** side.
[0092] The shaft coupling **58** on the driven side is also referred to as a coupling on a container side. In addition, the shaft coupling **19** on the drive side is also referred to as a coupling on a mounting portion side.
[0093] As shown in FIG. **6**, the shaft coupling **58** on the driven side has a connection portion **581** and a transmission shaft portion **582**.
[0094] The connection portion **581** is a portion in which a plurality of projection portions are provided on a circular main body having an outer diameter smaller than an inner diameter of the second internal space S**2**. The transmission shaft portion **582** is a shaft that protrudes along the upstream side in the mounting direction D**1** from a center of an end portion of the connection portion **581** on the upstream side in the mounting direction D**1**.
[0095] The shaft coupling **58** on the driven side is attached in a state of being rotatably supported by an attachment hole provided to pass through a central portion of the partition wall **532**. In this case, the transmission shaft portion **582** is disposed to pass through the central portion of the partition wall **532** and enter the first internal space S**1**.
[0096] The second internal space S**2** is provided with a covering portion **533** that covers and protects a periphery of (the connection portion **581** of) the shaft coupling **58** on the driven side is provided in the second internal space S**2**.
[0097] The covering portion **533** is formed as a structural portion that protrudes from the central portion of the partition wall **532** in the mounting direction D**1**. Specifically, the covering portion **533** is formed as a structural portion that is curved in a cylindrical shape with an opening at an upper portion.
Discharge Port
[0098] In addition, as shown in FIG. **6** and FIG. **7**A, the end portion holding body **53** is provided with the discharge port **55** in a lower section of the first internal space S**1** of the main body portion **531** to discharge the accommodated developer G (toner T).
[0099] The discharge port **55** in the first exemplary embodiment is formed as an opening having an approximately rectangular opening shape and extending from an inside of the first internal space S**1** to an outer peripheral surface (outer peripheral portion) of the main body portion **531**. In addition, the discharge port **55** is provided in a form in which one side surface (inner wall) thereof is formed

of a part of the partition wall **532** (see FIG. **6**).

[0100] The discharge port **55** is opened and closed from the outside by an opening and closing portion **56**.

Opening and Closing Portion of Discharge Port

[0101] The opening and closing portion **56** is an opening and closing lid that is slidably moved to open and close the discharge port **55**. The opening and closing portion **56** is also referred to as a shutter.

[0102] The opening and closing portion **56** is present at a position to close the discharge port **55** when the developer container **5** is not mounted in the mounting portion **15**, or when the developer container **5** is being mounted in the mounting portion **15** or is being detached from the mounting portion **15**. On the other hand, the opening and closing portion **56** is present at a position to open the discharge port **55** immediately before the developer container **5** is mounted in the mounting portion **15** or when the mounting is completed.

[0103] The opening and closing portion **56** in the first exemplary embodiment is configured as a rectangular plate-shaped lid. As shown in FIG. **7**A and the like, the opening and closing portion **56** has a rectangular plate-shaped opening and closing main body **561***a* and attachment side wall portions **561***b*. The attachment side wall portions **561***b* are portions that rise upward from both end portions of the opening and closing main body **561***a* along opening and closing directions E**1** and E**2** described below and are attached to side wall portions **572** of an attachment portion **57** described below.

[0104] In addition, as shown in FIG. **6**, the opening and closing portion **56** is provided with a sealing (seal) material **562** on a surface (inner surface) facing the discharge port **55**, the sealing material **562** covering the discharge port **55** when the opening and closing portion **56** is moved to the position to close the discharge port **55** and sealing the discharge port **55** to prevent leakage of the developer G (toner T).

[0105] Furthermore, as shown in FIG. **6** and the like, the opening and closing portion **56** is provided with a peripheral sealing sheet **563** on the inner surface, the peripheral sealing sheet **563** surrounding the sealing material **562**. The peripheral sealing sheet **563** has a shape having a protruding portion **563***c* that protrudes toward the downstream side in the mounting direction D**1** from the opening and closing portion **56**.

Attachment Portion of Opening and Closing Portion

[0106] Furthermore, as shown in FIGS. **5**, **6**, and **7**A, the end portion holding body **53** is provided with the attachment portion **57** to which the opening and closing portion **56** is movably attached.

[0107] The attachment portion **57** is a plate-like structural portion that is formed to surround the discharge port **55** on the outer peripheral surface (outer peripheral portion) of the main body portion **531**.

[0108] The attachment portion **57** in the first exemplary embodiment is configured as a structural portion to which the opening and closing portion **56** is attached to be movable in directions indicated by the arrows E**1** and E**2** along the mounting direction D**1**.

[0109] The direction indicated by the arrow E**1** is a movement direction when the opening and closing portion **56** opens the discharge port **55** with respect to the attachment portion **57**. In addition, the direction indicated by the arrow E**1** corresponds to a direction opposite to the mounting direction D**1**. On the other hand, the direction indicated by the arrow E**2** is a movement direction when the opening and closing portion **56** closes the discharge port **55** with respect to the attachment portion **57**.

[0110] In addition, as shown in FIG. **7**A and the like, the attachment portion **57** has a plate-shaped attachment main body **571** having a rectangular shape surrounding the discharge port **55**, and the side wall portions **572** that hang downward from both end portions of the attachment main body **571** along the opening and closing directions E**1** and E**2** of the opening and closing portion **56**.

[0111] The attachment main body **571** is provided in a state of protruding outward from a lower outer peripheral surface of the cylindrical main body portion **531** of the end portion holding body **53**. A surface of the attachment main body **571** to which the opening and closing portion **56** is attached is formed as a substantially flat surface portion that surrounds the discharge port **55** except for an opening portion corresponding to the discharge port **55** (see FIG. **7**B).

[0112] As shown in FIG. **5** and the like, the opening and closing portion **56** is attached in a state where the side wall portion **561***b* thereof is movably caught and held by the side wall portion **572** of the attachment portion **57**.

[0113] In addition, as shown in FIG. **5** and the like, the end portion holding body **53** is provided with a depressed portion **534** that is depressed inwardly in a portion of the cylindrical main body portion **531** that is located in an upper section of the second internal space S**2**.

[0114] A storage medium **59** that stores information regarding the developer container **5** is disposed in the depressed portion **534**. In a case where the developer container **5** is mounted in the mounting portion **15**, the storage medium **59** is electrically connected to a connection terminal (not shown) provided on the mounting portion **15** side to read and write the information.

[0115] In the first exemplary embodiment, the depressed portion **534** has a structure in which a lower surface thereof is held from below as both end portions of the opening in the covering portion **533** of the shaft coupling **58** on the driven side are connected to the lower surface (see FIG. **7** and the like).

[0116] In addition, as shown in FIGS. **4**, **5**, and the like, the end portion holding body **53** is provided with a rotation holding portion **537** in an outer peripheral portion of the cylindrical main body portion **531** that is located in an upper section of the first internal space S**1**.

[0117] The rotation holding portion **537** is a portion that rotatably holds the agitating body **54**.

[0118] A plurality of the rotation holding portions **537** in the first exemplary embodiment are provided in the main body portion **531** at intervals in a circumferential direction of the outer peripheral portion that is located in the upper section of the first internal space S**1**. In addition, the rotation holding portion **537** is provided as a projection portion having a shape protruding toward the first internal space S**1** in an outer peripheral portion thereof.

[0119] In addition, as shown in FIGS. **5**, **7**, and the like, the end portion holding body **53** is provided with a cutout portion **535** in a portion of the main body portion **531** that is located in a lower section of the second internal space S**2**.

[0120] The cutout portion **535** is a portion cut out so as to enter the corresponding portion of the main body portion **531** on a side (inside) opposite to the mounting direction D**1**.

[0121] When the developer container **5** is mounted in the mounting portion **15**, the cutout portion **535** is formed such that a mounting fitting portion **175**, which will be described later, is fitted into a depressed edge portion of the cutout portion **535** (see FIGS. **9** and **10**).

[0122] In addition, as shown in FIG. **7** and the like, the end portion holding body **53** is provided with a pair of guide receiving portions **538** and **539** on the outer peripheral portion of the main body portion **531**.

[0123] The guide receiving portions **538** and **539** are portions that are guided by being placed on a pair of guide portions **162** and **163**, which will be described later, in the mounting platform **16** of the mounting portion **15**, during insertion and removal operations when the developer container **5** is attached to and detached from the mounting portion **15**.

[0124] Furthermore, as shown in FIGS. **5**, **6**, and the like, the end portion holding body **53** is provided with an identification shape portion **70** in the outer peripheral portion of the cylindrical main body portion **531** that is located in the upper section of the first internal space S**1**.

[0125] The identification shape portion **70** is a portion formed as an identification shape to distinguish whether or not the developer container **5** can be mounted according to differences in types of the developer and the accommodation portion **51**. The identification shape portion **70** is also referred to as, for example, a mechanical key.

[0126] The identification shape portion **70** in the first exemplary embodiment is a portion in which identification projection portions **71** and identification recess portions **72** are combined on the outer peripheral portion of the main body portion **531** in the circumferential direction. The identification projection portion **71** and the identification recess portion **72** are formed as a projection portion and a recess portion having predetermined lengths in the circumferential direction. The identification recess portion **72** is also a portion formed between the plurality of identification projection portions **71**.

[0127] The identification shape portion **70** is formed to have a different uneven pattern corresponding to the difference in the type of the developer or the type of the accommodation portion **51**. Therefore, the identification shape portions **70** in the developer containers **5**A, **5**B, **5**C, and **5**D are formed in different uneven patterns, for example, due to the difference in the type of the developer.

Agitating Body

[0128] The agitating body **54** is a structure attached to the open end portion **52***a* of the accommodation main body **52**.

[0129] In addition, the agitating body **54** is a structure having a function of agitating the developer G (toner T) that moves from an inside of the accommodation main body **52** is discharged through the open end portion **52***a*.

[0130] As shown in FIGS. **4** and **6**, and the like, the agitating body **54** in the first exemplary embodiment has a cylindrical main body portion **541** of which both end portions **541***a* and **541***b* are open.

[0131] As shown in FIG. **6** and the like, the main body portion **541** is configured so that an entire portion including a shaft fitting portion **545** and an agitating portion **546**, which will be described later, is accommodated in the first internal space S**1** of the end portion holding body **53**.

[0132] In addition, as shown in FIG. **6** and the like, the main body portion **541** is configured so that the coupled portion **523** of the accommodation main body **52** is fitted into an internal space of the end portion **541***b* on the upstream side in the mounting direction D**1**. An inner wall of the end portion **541***b* of the main body portion **541** on the upstream side is formed with a coupling portion **543** (see FIG. **6**) to be coupled to the coupled portion **523** of the accommodation main body **52**.

[0133] Furthermore, the main body portion **541** is provided with an annular flange portion **544** that continuously protrudes in the circumferential direction on an outer peripheral portion of the end portion **541***a* on the downstream side in the mounting direction D**1**. The flange portion **544** is rotatably held by the rotation holding portion **537** of the end portion holding body **53** when the main body portion **541** is fitted into the first internal space S**1** of the end portion holding body **53** (see FIG. **7** and the like).

[0134] In addition, as shown in FIGS. **4**, **6**, and the like, the agitating body **54** is provided with the shaft fitting portion **545** at the end portion **541***a* of the main body portion **541** on the downstream side in the mounting direction D**1**.

[0135] The shaft fitting portion **545** is formed as a cylindrical structural portion that protrudes in the mounting direction D**1** from a central portion of a support plate that is fixed to extend inside the opening of the end portion **541***a* in a diameter direction thereof. As shown in FIG. **6**, the shaft fitting portion **545** is provided with a fitting hole **545***a* into which the transmission shaft portion **582** of the shaft coupling **58** on the driven side is fitted and fixed, at a center portion thereof.

[0136] Furthermore, as shown in FIGS. **4**, **6**, and the like, the agitating body **54** is provided with the agitating portion **546** at the end portion **541***a* of the main body portion **541** on the downstream side in the mounting direction D**1**.

[0137] The agitating portion **546** is formed as two rod-shaped structural portions protruding in the mounting direction D**1** from both side portions of the support plate with the shaft fitting portion **545** interposed therebetween.

[0138] All of the developer containers **5**A, **5**B, **5**C, and **5**D have a structure formed by assembling the accommodation portion **51** in which the portion to which the agitating body **54** of the

accommodation main body **52** is attached is rotatably held by the end portion holding body **53**.
Mounting platform of Mounting Portion
[0139] As shown in FIG. **8**, the respective mounting platforms **16** in the mounting portions **15**A, **15**B, **15**C, and **15**D have plate-shaped holding platforms **161** elongated in one direction to respectively hold lower portions of the developer containers **5**A, **5**B, **5**C, and **5**D.
[0140] The holding platform **161** is provided with side walls that rise upward at both end portions along a longitudinal direction thereof. Upper ends of the side walls are formed as guide portions **162** and **163**. The guide portions **162** and **163** are portions that guide the developer containers **5**A, **5**B, **5**C, and **5**D along the mounting direction D**1** and the detachment direction D**2** during the insertion and removal for the attachment and detachment, respectively, and rotatably hold the accommodation main body **52** of the accommodation portion **51**.
[0141] The holding platform **161** is provided with a termination wall **164** at an end portion in the mounting direction D**1**.
[0142] A holding recess portion **164***a* that introduces and holds the covering portion **533** of the end portion holding body **53** of the developer containers **5**A, **5**B, **5**C, and **5**D is provided on an upper portion of the termination wall **164**. The holding recess portion **164***a* is formed as a portion having a shape depressed in an arc shape.
[0143] In addition, the holding platform **161** is provided with a passage port **165** through which the developer passes in a portion of a bottom surface portion thereof on an inner side in the mounting direction D**1**. The passage port **165** is provided at a position facing the discharge port **55** of each of the mounted developer containers **5**A, **5**B, **5**C, and **5**D.
[0144] Furthermore, the holding platform **161** is provided with an opening and closing guide body **170** in a portion on the inner side in the mounting direction D**1** to surround the passage port **165** in the bottom surface portion thereof.
[0145] The opening and closing guide body **170** includes a placing platform **171** for the attachment portion **57**, an opening and closing portion holding portion **173** that holds the opening and closing portion **56**, and a mounting fitting portion **175**.
[0146] The placing platform **171** is formed as a pair of flat surface platforms parallel to each other along the mounting direction D**1** at an interval to secure a facing space S**3**.
[0147] The facing space S**3** is a space facing the discharge port **55** in the developer containers **5**A, **5**B, **5**C, and **5**D and the passage port **165** in the holding platform **161**. On the placing platform **171**, a portion of the attachment portion **57** in the developer containers **5**A, **5**B, **5**C, and **5**D that is not present when the opening and closing portion **56** is moved in a direction of opening the discharge port **55** is placed.
[0148] The opening and closing portion holding portion **173** is formed as a pair of holding arm portions **173**A and **173**B that are parallel to each other along the mounting direction D**1** at an interval to secure a holding space S**4**.
[0149] The holding space S**4** is a space for holding the opening and closing portions **56** in the developer containers **5**A, **5**B, **5**C, and **5**D from both sides along the mounting direction D**1** when the opening and closing portion **56** is mounted. The opening and closing portion holding portion **173** holds and stops the opening and closing portion **56** in the middle of the mounting of the opening and closing portion **56**, and moves the opening and closing portion **56** in the movement direction E**1** when the opening and closing portion **56** is opened with respect to the attachment portion **57**.
[0150] The mounting fitting portion **175** is a portion that is fitted into the cutout portion **535** of the end portion holding body **53** of the developer container **5** when the developer container **5** is mounted in the mounting portion **15** (see FIGS. **9** and **10**).
[0151] The mounting fitting portion **175** also serves as a portion that connects and holds the pair of flat surface platforms of the placing platform **171**.
Mounting of Developer Container in Mounting Portion

[0152] In a case where the developer containers **5**A, **5**B, **5**C, and **5**D are respectively mounted in the mounting portions **15**A, **15**B, **15**C, and **15**D, the developer containers **5**A, **5**B, **5**C, and **5**D are guided in the mounting direction D**1** by the guide portions **162** and **163** of the holding platform **161** and are pushed into the respective storage spaces.
[0153] In addition, during the pushing of the developer containers **5**A, **5**B, **5**C, and **5**D, the opening and closing portion **56** is held and stopped by the opening and closing portion holding portion **173** of the opening and closing guide body **170**, and then the opening and closing portion **56** is moved in the movement direction E**1** when the opening and closing portion **56** is opened with respect to the attachment portion **57**.
[0154] Subsequently, the developer containers **5**A, **5**B, **5**C, and **5**D enter a state where the mounting fitting portion **175** of the opening and closing guide body **170** is fitted into the cutout portion **535** of the end portion holding body **53**.
[0155] Accordingly, the mounting of the developer containers **5**A, **5**B, **5**C, and **5**D in the respective mounting portions **15**A, **15**B, **15**C, and **15**D is completed.
[0156] In a process of completing the mounting of the developer containers **5**A, **5**B, **5**C, and **5**D, the opening and closing portion **56** is moved in the movement direction E**1** in which the opening and closing portion **56** is opened to open the discharge port **55**. In this case, the discharge port **55** enters a state of facing and being continuous with the passage port **165** of the mounting platform **16** through the facing space S**3**.
[0157] In addition, the shaft coupling **58** on the driven side in the developer containers **5**A, **5**B, **5**C, and **5**D is connected to the shaft coupling **19** on the drive side in the mounting portions **15**A, **15**B, **15**C, and **15**D. Accordingly, in a case where the rotational power is transmitted to the shaft coupling **58** on the driven side, the developer containers **5**A, **5**B, **5**C, and **5**D are rotated in the direction of the arrow M in a state where the accommodation main body **52** and the agitating body **54** included in the accommodation portion **51** are held by the end portion holding body **53**.
[0158] In a process of mounting the developer containers **5**A, **5**B, **5**C, and **5**D, the identification shape portion **70** distinguishes whether or not the developer containers **5**A, **5**B, **5**C, and **5**D can be mounted with an identification matching portion (not shown) provided at each of corresponding positions of the storage spaces of the mounting portions **15**A, **15**B, **15**C, and **15**D.
[0159] In this case, the developer containers **5**A, **5**B, **5**C, and **5**D can be mounted in the mounting portions **15** only in a case where the identification shape portion **70** matches the identification matching portion. In contrast, in a case where the identification shape portion **70** does not match the identification matching portion and is inconsistent, the developer containers **5**A, **5**B, **5**C, and **5**D cannot be mounted in the mounting portion **15**, and erroneous mounting is prevented.
[0160] When it is time to replenish the developer, the developer containers **5**A, **5**B, **5**C, **5**D, which have been mounted, are rotated by a predetermined amount in the direction of the arrow M in a state where the accommodation main body **52** and the agitating body **54** are held by the end portion holding body **53**.
[0161] Accordingly, the developer G (toner T) accommodated in the accommodation main body **52** is transported to be moved to an agitating body **54** side by the helical ridge portion **525** of the rotating accommodation main body **52**. Subsequently, the developer G (toner T) transported in advance is sequentially discharged and falls from the discharge port **55** provided in a lower portion of the end portion holding body **53**. In a case where the developer G (toner T) passes through the agitating body **54**, the developer G is agitated and broken up by the agitating portion **546**.
[0162] The developer G (toner T) discharged from the discharge port **55** passes through the passage port **165** in the mounting portions **15**A, **15**B, **15**C, and **15**D and is sent to the replenishing device **17**.
Further Configuration of Developer Container and the Like
[0163] In the developer containers **5**A, **5**B, **5**C, and **5**D, from the viewpoint of securing sealability of the discharge port **55** by the opening and closing portion **56** of the attachment portion **57**, a

plate-shaped extending portion **575** that extends from the attachment main body **571** toward the downstream side in the mounting direction D**1** is provided.

[0164] The extending portion **575** is an extension of an end portion of the plate-shaped attachment main body **571** on the downstream side in the mounting direction D**1** away from the discharge port **55** toward the downstream side in the mounting direction D**1**.

[0165] By providing the extending portion **575** in the attachment portion **57**, a planar surface of the attachment main body **571** on which the opening and closing portion **56** is present is extended to be wide toward the downstream side of the discharge port **55** in the mounting direction D**1**. That is, regarding the attachment portion **57**, the attachment main body **571** has a wider flat surface portion.

[0166] As a result, as shown in FIG. **7**A and the like, when the opening and closing portion **56** enters a state of closing the discharge port **55**, the attachment portion **57** can be brought into a state of facing the opening and closing portion **56** with a large margin to the extending portion **575** of the attachment main body **571**.

[0167] Accordingly, as shown in FIG. **7**A and the like, the opening and closing portion **56** in the first exemplary embodiment can particularly bring the protruding portion **563***c* of the peripheral sealing sheet **563** into wide contact with the flat surface portion of the extending portion **575**, thereby improving the sealability of the discharge port **55**.

[0168] In addition, as shown in FIGS. **5**, **7**, and the like, in the developer containers **5**A, **5**B, **5**C, and **5**D, three support portions **6** that support the extending portion **575** are provided on an extending surface of the extending portion **575** of the attachment portion **57** on a side opposite to the opening and closing portion **56**.

[0169] The three support portions **6** include a first support portion **6**A, a second support portion **6**B, and a third support portion **6**C.

[0170] The first support portion **6**A is provided at a substantially central portion in a width direction perpendicular to the mounting direction D**1** on the surface of the extending portion **575** on the opposite side.

[0171] In addition, the first support portion **6**A is formed as a plate-shaped portion **60** (see FIG. **5**) that is substantially orthogonal to the surface of the extending portion **575** on the opposite side.

[0172] In addition, the first support portion **6**A is provided in a form in which the plate-shaped portion **60** is connected and fixed to the partition wall **532** of the end portion holding body **53**.

[0173] Furthermore, the first support portion **6**A is provided in a form connected to the covering portion **533** of the end portion holding body **53**, which is an example of an opposing structural portion having a portion facing the extending portion **575**.

[0174] Specifically, the first support portion **6**A is connected to abut against a curved surface of a lower portion of the covering portion **533** curved in an arc shape.

[0175] The second support portion **6**B and the third support portion **6**C are respectively provided in portions slightly inner than both end portions in the width direction orthogonal to the mounting direction D**1** on the surface of the extending portion **575** on the opposite side. In other words, the second support portion **6**B and the third support portion **6**C are provided to be present at positions at an interval on both sides of the first support portion **6**A with the first support portion **6**A interposed therebetween.

[0176] In addition, the second support portion **6**B and the third support portion **6**C are formed as plate-shaped portions (see FIG. **5**) substantially orthogonal to the surface of the extending portion **575** on the opposite side, similarly to the first support portion **6**A.

[0177] Furthermore, the second support portion **6**B and the third support portion **6**C are provided in a form in which one end portion (upper end portion) thereof is connected to a part of the end portion holding body **53**, which is the opposing structural portion.

[0178] Specifically, the second support portion **6**B and the third support portion **6**C are connected to abut against a curved surface portion of a lower portion on an inner side of the cutout portion **535** of the end portion holding body **53**.

[0179] Therefore, according to the developer containers **5**A, **5**B, **5**C, and **5**D, regarding the extending portion **575**, compared to a case where the extending portion **575** is not provided with the support portion **6** that supports the extending portion **575**, the extending portion **575** can become a portion that is less likely to be deformed or damaged by providing the flat surface portion that is made wider to secure the sealability of the discharge port **55** by the opening and closing portion **56**.
[0180] In addition, in the developer containers **5**A, **5**B, **5**C, and **5**D according to the first exemplary embodiment, three support portions **6**, that is, the first support portion **6**A, the second support portion **6**B, and the third support portion **6**C are provided as the support portion **6**.
[0181] Therefore, regarding the extending portion **575**, compared to a case where one support portion or two support portions are provided as the support portion **6**, the extending portion **575** can become a portion that is less likely to be deformed or damaged by providing the flat surface portion that is made even wider in the opening and closing portion **56**.
[0182] Incidentally, regarding the extending portion **575**, for example, it is also considered to increase a thickness of the extending portion **575** instead of providing the support portion **6**.
[0183] However, in this case, when the extending portion **575** is produced by a plastic molding method, depression-like deformation (sink mark) may occur due to molding, and a smooth surface of a surface to which the opening and closing portion **56** is attached may not be obtained. In this case, the sealability in the extending portion **575** of the attachment portion **57** deteriorates, and leakage of the developer may be incurred.
[0184] In this regard, in the extending portion **575** on which the support portion **6** is provided, the extending portion **575** is less likely to be deformed due to sink marks or the like.
[0185] In addition, when the developer containers **5**A, **5**B, **5**C, and **5**D are mounted in the mounting portion **15** or handled by holding the grip portion **527** and the like by hand, the user may cause a lower portion of a front end in the mounting direction D**1** to be brought into contact with a surrounding portion of the mounting portion **15** or to be dropped and hit by an object present at a place where the lower portion of the front end is dropped.
[0186] That is, in a case of handling the developer containers **5**A, **5**B, **5**C, and **5**D, the developer containers **5**A, **5**B, **5**C, and **5**D may be erroneously operated by the user as described above.
[0187] Incidentally, in the developer containers **5**A, **5**B, **5**C, and **5**D, there may be a case where the developer G is unevenly present on one end portion side of either of both ends in the longitudinal direction, and a case where an overall weight balance becomes uneven due to the reason that the developer containers **5**A, **5**B, **5**C, and **5**D have the container structure having the end portion holding body **53**.
[0188] Therefore, the front end of the developer containers **5**A, **5**B, **5**C, and **5**D in the mounting direction D**1** or a rear end opposite to the front end tends to be directed downward, and a probability of falling from the front end or the rear end increases. This may easily induce erroneous operation by the user mentioned above.
[0189] In a case where an erroneous operation or the like by the user occurs, the extending portion **575** that is present to extend on the front end side of the mounting direction D**1** of the developer containers **5**A, **5**B, **5**C, and **5**D is more likely to come into contact with or collide with another object, increasing a probability of damage.
[0190] In this regard, in the extending portion **575** provided with the first support portion **6**A, the second support portion **6**B, and the third support portion **6**C, strength of the extending portion **575** is increased by each of the support portions **6**A, **6**B, and **6**C, and thus the extending portion **575** is less likely to be damaged.
[0191] In addition, in the developer containers **5**A, **5**B, **5**C, and **5**D, the first support portion **6**A is provided in a form connected to the covering portion **533**, and the second support portion **6**B and the third support portion **6**C are provided in a form connected to a part of the end portion holding body **53**.

[0192] Therefore, according to the developer containers **5**A, **5**B, **5**C, and **5**D, compared to a case where the first support portion **6**A, the second support portion **6**B, and the third support portion **6**C are provided in a form not connected to the covering portion **533** or the part of the end portion holding body **53**, the strength of the extending portion **575** is further increased by each of the support portions **6**A, **6**B, and **6**C.
[0193] Accordingly, the extending portion **575** is less likely to be deformed or damaged.
[0194] In addition, compared to a case where the connection is not provided, it is more difficult for a user's finger or the like to enter a gap between the extending portion **575** and the covering portion **533** or a gap between the extending portion **575** and the part of the end portion holding body **53**. In short, the presence of any one of the support portions **6**A, **6**B, and **6**C in a part of each of the gaps may prevent entrance of the user's finger or the like.
[0195] Moreover, a probability of the occurrence of deformation of the extending portion **575** and deformation of the covering portion **533** or the part of the end portion holding body **53** to which the support portion **6** is connected in a case where the user's finger or the like enters each of the gaps is avoided or reduced.
[0196] In addition, in the developer containers **5**A, **5**B, **5**C, and **5**D, as shown in FIG. **5** and the like, the first support portion **6**A is connected to a portion of an arcuate curved surface shape that is convex downward in the covering portion **533**, which is the example of the opposing structural portion.
[0197] Therefore, according to the developer containers **5**A, **5**B, **5**C, and **5**D, compared to a case where the first support portion **6**A is connected to a flat surface-shaped portion in the opposing structural portion, even in a case where an external force F such as an impact or vibration is applied to the extending portion **575** through the plate-shaped portion **60** of the first support portion **6**A, the external force F is easily distributed and transmitted as, for example, forces fa and fb in a curved surface-shaped portion of the covering portion **533** (see FIG. **5**). Accordingly, the extending portion **575** is less likely to be deformed or damaged.
[0198] Next, in the exemplary embodiment, as shown in FIGS. **10** and **11**, the first support portion **6**A is configured as a portion that is fitted into a recess portion **176** in the mounting portion **15**.
[0199] Specifically, at least the plate-shaped portion **60** of the first support portion **6**A is fitted into the recess portion **176**.
[0200] In this case, as shown in FIG. **9** and the like, in the mounting portions **15**A, **15**B, **15**C, and **15**D, the recess portion **176** is provided in the mounting fitting portion **175** of the opening and closing guide body **170**.
[0201] The recess portion **176** is formed by disposing a pair of projection portions **177**A and **177**B, which protrude from a side surface **175***a* of the mounting fitting portion **175** facing the developer container **5** to be mounted toward the side opposite to the mounting direction D**1**, at a predetermined interval. A groove width of the recess portion **176** is set to be slightly larger than a thickness of the plate-shaped portion **60** of the first support portion **6**A.
[0202] When the developer containers **5**A, **5**B, **5**C, and **5**D are mounted in the mounting portions **15**A, **15**B, **15**C, and **15**D, as shown in FIGS. **10** and **11**B, the plate-shaped portion **60** of the first support portion **6**A enters a state of being fitted into the recess portion **176** of the mounting fitting portion **175**.
[0203] Therefore, according to the developer containers **5**A, **5**B, **5**C, and **5**D, compared to a case where the plate-shaped first support portion **6**A is not configured as a portion that is fitted into the recess portion **176** provided in the mounting portion **15** as a mounting destination during the mounting, the plate-shaped first support portion **6**A is kept in a state where the plate-shaped first support portion **6**A is less likely to move due to the recess portion **176**.
[0204] Here, in a stage of mounting the developer container **5** in the mounting portion **15** or after the mounting, the accommodation portion **51** (the accommodation main body **52** or the like) of the developer container **5** may be subjected to an unexpected movement due to an unnecessary

operation of the user or to vibration caused by an operation of the developer container **5** or the like. Arrows J**1** and J**2** in FIG. **11**B indicate directions in which the end portion holding body **53** may move when subjected to a movement that causes rotation due to an unnecessary operation by the user.

[0205] That is, even at such times, as long as the developer containers **5**A, **5**B, **5**C, and **5**D are used, even in a case where the developer containers **5**A, **5**B, **5**C, and **5**D are subjected to external forces due to the movement and vibration, the plate-shaped first support portion **6**A is kept in a state where the plate-shaped first support portion **6**A is less likely to move due to the recess portion **176**. Therefore, the unnecessary external forces are less likely to be applied to the extending portion **575** through the plate-shaped first support portion **6**A. As a result, the extending portion **575** is less likely to be deformed or damaged.

[0206] Next, in the exemplary embodiment, as shown in FIG. **5** and the like, the first support portion **6**A is provided with a protruding portion **61** that protrudes toward the front end side with respect to an end portion **575***a* (see FIG. **7**B and the like) of the extending portion **575***b* on the front end side in the mounting direction D**1**.

[0207] The protruding portion **61** in the first exemplary embodiment is provided in a plate-shaped form that protrudes toward the downstream side in the mounting direction D**1** from an upper portion of the plate-shaped portion **60** of the first support portion **6**A. In addition, the protruding portion **61** is provided in a state where the upper portion thereof is connected to the covering portion **533**.

[0208] Therefore, according to the developer containers **5**A, **5**B, **5**C, and **5**D, compared to a case where the protruding portion **61** is not provided in the first support portion **6**A, even in a case where the extending portion **575** is subjected to an impact from the outside in the event of a fall, collision, or the like, the impact is more likely to be received by the protruding portion **61** of the first support portion **6**A than by the extending portion **575**. As a result, a situation in which the extending portion **575** is subjected to the impact from the outside first can be reduced. Accordingly, the extending portion **575** is less likely to be deformed or damaged.

[0209] Moreover, as shown in FIG. **12** and the like, the protruding portion **61** of the first support portion **6**A in the first exemplary embodiment is provided in a form that is present on the upstream side in the mounting direction D**1** with respect to an end portion **533***e* of a portion of the covering portion **533** facing the first support portion **6**A on the downstream side in the mounting direction D**1**.

[0210] The protruding portion **61** at this time is provided in a form that protrudes toward the downstream side in the mounting direction D**1** with respect to the end portion **575***a* of the extending portion **575** on the downstream side in the mounting direction D**1**.

[0211] Therefore, according to the developer containers **5**A, **5**B, **5**C, and **5**D, compared to a case where the protruding portion **61** of the first support portion **6**A is provided in a form that protrudes toward the downstream side in the mounting direction D**1** with respect to the end portion **533***e* of the covering portion **533**, a situation in which the protruding portion **61** is subjected to an unnecessary collision or impact from the downstream side in the mounting direction D**1** before the end portion **533***e* of the covering portion **533** can be suppressed.

[0212] Accordingly, the extending portion **575** is less likely to be deformed or damaged.

[0213] In addition, the protruding portion **61** of the first support portion **6**A is provided on an upper portion side of the plate-shaped portion **60**. In other words, the protruding portion **61** is provided at a portion away from the discharge port **55** compared to a case where the protruding portion **61** is provided on a lower portion side of the plate-shaped portion **60**.

[0214] Therefore, even when the developer containers **5**A, **5**B, **5**C, and **5**D fall or the like due to an erroneous operation of the user, a probability that the protruding portion **61** of the first support portion **6**A is subjected to an impact (external force F) is higher than that in a case where the protruding portion **61** is provided on the lower portion side of the plate-shaped portion **60**.

[0215] Accordingly, in the developer containers **5**A, **5**B, **5**C, and **5**D, even in a case where the developer containers **5**A, **5**B, **5**C, and **5**D fall or the like, the extending portion **575** and other portions (the discharge port **55** and the like) of the attachment portion **57** are less likely to be deformed or damaged.
[0216] In the exemplary embodiment, the first support portion **6**A can be configured to be used as an identification shape portion **80**.
[0217] As in the case of the identification shape portion **70**, the identification shape portion **80** in this case is a portion having a shape to distinguish whether or not the developer container **5** can be mounted in the mounting portion **15** according to a difference in the type of the developer G or a difference in the type of the accommodation portion **51**.
[0218] The identification shape portion **80** of the first support portion **6**A is configured with, for example, a positional relationship or a dimensional difference between the plate-shaped portion **60** and the protruding portion **61** in the first support portion **6**A, or a projection portion or a recess portion provided in each of the plate-shaped portion **60** and the protruding portion **61**.
[0219] In the first support portion **6**A according to the first exemplary embodiment, the protruding portion **61** is used as the identification shape portion **80**, and the mounting portion **15** is provided with a matching recess portion **178** as an example of a matching shape portion into which the protruding portion **61** is fitted as shown in FIG. **11**A. As shown in FIGS. **9**, **11**A, and the like, the matching recess portion **178** is formed by providing a predetermined gap in a substantially central portion of the mounting fitting portion **175**.
[0220] In this case, different types of the developer containers **5** may be accommodated in a manner in which the protruding portions **61** having different shapes are provided in the first support portions **6**A, while the mounting portions **15** corresponding to the developer containers **5** are either configured without the matching recess portion **178** or provided with matching recess portions having shapes that match the protruding portions **61** having different shapes.
[0221] Therefore, according to the developer containers **5**A, **5**B, **5**C, and **5**D, compared to a case where the first support portion **6**A is not configured to be used as the identification shape portion **80**, the first support portion **6**A can also be used as a portion having a function of distinguishing whether or not the developer container **5** can be mounted in the mounting portion **15** according to differences in the types of the developer G and the accommodation portion **51**. That is, the first support portion **6**A can be effectively used.
[0222] In addition, according to the developer containers **5**A, **5**B, **5**C, and **5**D, even in a case where the number of types of the developer and the accommodation portion is increased and it is difficult to cope with the increase with only the above-described identification shape portion **70**, the first support portion **6**A is configured to be used as the identification shape portion **80**, whereby it is possible to secure a function of distinguishing whether or not the various types of developer containers **5** can be mounted in the mounting portion **15**.
[0223] Furthermore, in the developer containers **5**A, **5**B, **5**C, and **5**D, as shown in FIG. **11**B, the second support portion **6**B and the third support portion **6**C present on both sides of the first support portion **6**A are configured to be respectively fitted into fitting recess portions **177**Ac and **177**Bc of one side surface portions of the pair of projection portions **177**A and **177**B in the opening and closing guide body **170**.
[0224] Even with this, during the mounting, the developer containers **5**A, **5**B, **5**C, and **5**D are kept from moving easily by the fitting between the second support portion **6**B and the third support portion **6**C and the fitting recess portions **177**Ac and **177**Bc.
[0225] Incidentally, the second support portion **6**B and the third support portion **6**C may also be configured as a part of the identification shape portion **80**.
Modification Example
[0226] The present invention is not limited to the configuration described as the first exemplary embodiment, and various modifications or improvements can be made without changing the gist of

each invention described in the claims. Therefore, the present invention also includes, for example, the following modification examples.
[0227] As shown in FIG. **13**, in the developer container **5** according to the exemplary embodiment of the present invention, as the support portion **6**, two support portions, a fourth support portions **6**D and a fifth support portion **6**E may be provided at an interval in the width direction of the extending portion **575**.
[0228] The fourth support portion **6**D is a support portion provided in a state of being connected to the covering portion **533**, similarly to the case of the first support portion **6**A. The fifth support portion **6**E is a support portion provided in a state of not being connected to the covering portion **533**. Both the fourth support portion **6**D and the fifth support portion **6**E are provided in a state of being connected to the partition wall **532** of the end portion holding body **53**.
[0229] In a case where the fourth support portion **6**D and the fifth support portion **6**E are provided as described above, each of the support portions **6**D and **6**E may be configured as a different type of identification shape portion **80**. In this case, the mounting portion **15** is also provided with a matching shape portion such as a recess portion that has a shape that matches the different type of identification shape portion **80** in each of the support portions **6**D and **6**E and enables the mounting of the accommodation portion **51** of the developer container **5**.
[0230] In such a case, even in a case where the number of types of the developer or types of the accommodation portion **51** of the developer container **5** is further increased, it is possible to prevent the occurrence of erroneous mounting of the developer container **5** in the mounting portion **15**.
[0231] In addition, as shown by a two-dot chain line in FIG. **13**, the developer container **5** in the exemplary embodiment of the present invention may be provided with an additional identification shape portion **85** at an inner part of the end portion holding body **53** of the accommodation portion **51** on the front end side in the mounting direction D**1**. In this case, the mounting portion **15** is also provided with a matching shape portion such as a recess portion that has a shape that matches the additional identification shape portion **85** and enables the mounting of the accommodation portion **51** of the developer container **5**.
[0232] With this configuration, even though the number of types of the developer or types of the accommodation portion **51** of the developer container **5** is further increased, it is possible to prevent the occurrence of erroneous mounting of the developer container **5** in the mounting portion **15**.
[0233] The support portion **6** in the developer container **5** in the exemplary embodiment of the present invention is not limited to a plate-shaped form, and may be a form consisting of a shape other than the plate shape as long as at least the extending portion **575** can be supported. In a case where a plurality of support portions **6** are provided, the number thereof is not limited to three, and may be, for example, two or four.
[0234] In addition, the support portion **6** may be provided to be connected to a two-surface shape (for example, a two-surface shape that is developed in a V shape) that is developed such that the portion of the opposing structural portion facing the extending portion **575** crosses diagonally.
[0235] The number of support portions **6** is not particularly limited, and may be, for example, one or two.
[0236] The accommodation portion **51** of the developer container **5** in the exemplary embodiment of the present invention is not limited to a rotary type accommodation portion in which the accommodation main body **52** and the like shown in the first exemplary embodiment are rotatably held by the end portion holding body **53**. As another type of accommodation portion **51**, for example, a fixed type accommodation portion may be disposed in which the accommodation portion **51** itself does not rotate and a transport member that rotates and transports the developer inside the accommodation portion **51**.
[0237] Even in the case of the fixed type accommodation portion **51**, in a case where a plate-shaped extending portion that extends toward the front end side in the mounting direction D**1** is provided in the attachment portion **57** of the opening and closing portion **56**, for example, the support

portion **6** may be provided.
[0238] In addition, the opening and closing portion **56** of the developer container **5** in the exemplary embodiment of the present invention may be an opening and closing portion that opens and closes the discharge port **55** by moving along an outer circumference of the cylindrical accommodation portion **51** of the fixed type in the circumferential direction. Even in this case, in a case where a plate-shaped extending portion that extends toward the front end side in the mounting direction D**1** is provided in the attachment portion **57** of the opening and closing portion **56**, for example, the support portion **6** may be provided.
[0239] Incidentally, in this case, a surface of a support body to which the opening and closing portion **56** of the attachment portion **57** is attached is a curved surface portion that is curved in an arc shape.
[0240] Furthermore, the image forming apparatus **1** using the developer container **5** in the exemplary embodiment of the present invention is not particularly limited to the number of developer containers **5** used, the number of colors in images, the configuration of the image forming section **20**, and the like.
Supplementary Notes
[0241] (((1)))
[0242] A developer container comprising:
[0243] an accommodation portion that accommodates a developer;
[0244] a discharge port through which the developer is discharged from an outer peripheral portion of the accommodation portion on a front end side in a mounting direction;
[0245] an opening and closing portion that opens and closes the discharge port from an outside; and
[0246] an attachment portion that surrounds the discharge port and to which the opening and closing portion is movably attached,
[0247] wherein the attachment portion has an extending portion extending toward the front end side in the mounting direction, and
[0248] the extending portion is provided with a support portion that supports the extending portion on a surface opposite to the opening and closing portion.
[0249] (((2)))
[0250] The developer container according to (((1))),
[0251] wherein the accommodation portion includes an opposing structural portion having a portion facing the extending portion at an inner part on the front end side in the mounting direction, and
[0252] the support portion is provided in a form connected to the opposing structural portion.
[0253] (((3)))
[0254] The developer container according to (((1))) or (((2))),
[0255] wherein the support portion has a protruding portion that protrudes toward the front end side with respect to an end portion of the extending portion on the front end side in the mounting direction.
[0256] (((4)))
[0257] The developer container according to any one of (((1))) to (((3)))),
[0258] wherein the support portion is provided with an identification shape portion for distinguishing whether or not the developer container can be mounted according to differences in types of the developer and the accommodation portion.
[0259] (((5)))
[0260] The developer container according to (((2))),
[0261] wherein the support portion has a plate shape, and
[0262] the plate-shaped support portion is configured as a portion that is fitted into a recess portion of a mounting destination when the accommodation portion is mounted.
[0263] (((6)))

[0264] The developer container according to (((2))),
[0265] wherein the support portion has a plate shape, and
[0266] the facing portion of the opposing structural portion has a curved surface shape of
[0267] which a projection side is connected to the plate-shaped support portion or has a two-surface shape that is developed to cross diagonally.
[0268] (((7)))
[0269] The developer container according to (((3))),
[0270] wherein the accommodation portion includes an opposing structural portion having a portion facing the extending portion at an inner part on the front end side in the mounting direction, and
[0271] the protruding portion is provided in a form that is present on an upstream side in the mounting direction with respect to an end portion of the facing portion on a downstream side in the mounting direction.
[0272] (((8)))
[0273] The developer container according to (((4))),
[0274] wherein the support portion is provided as a plurality of support portions at intervals, and
[0275] the identification shape portion is provided in each of the plurality of support portions.
[0276] (((9)))
[0277] The developer container according to any one of (((1))) to (((8))),
[0278] wherein the accommodation portion is provided with an additional identification shape portion for distinguishing whether or not the developer container can be mounted according to differences in types of the developer and the accommodation portion at an inner part on the front end side in the mounting direction.
[0279] (((10)))
[0280] An image forming apparatus comprising:
[0281] an image forming section that forms an image consisting of a developer; and a developer container that accommodates the developer supplied to the image forming section; and
[0282] a mounting portion in which the developer container is attachably and detachably mounted by inserting and removing the developer container,
[0283] wherein the developer container is the developer container according to any one of (((1))) to (((9))).
[0284] (((11)))
[0285] The image forming apparatus according to (((10))),
[0286] wherein the mounting portion is provided with a recess portion into which a support portion is fitted.
[0287] (((12)))
[0288] The image forming apparatus according to (((10))),
[0289] wherein, in a case where the developer container is the developer container according to (((4))),
[0290] the mounting portion is provided with an identification matching portion that has a shape that matches the identification shape portion and enables mounting of the accommodation portion.
[0291] (((13)))
[0292] The image forming apparatus according to (((10))),
[0293] wherein, in a case where the developer container is the developer container according to (((8))),
[0294] the mounting portion is provided with a plurality of identification matching portions that have shapes that respectively match the identification shape portions of the plurality of support portions and enable mounting of the accommodation portion.
[0295] (((14)))
[0296] The image forming apparatus according to (((10))),

[0297] wherein, in a case where the developer container is the developer container according to (((9))),
[0298] the mounting portion is provided with an additional identification matching portion that has a shape that matches the identification shape portion and enables mounting of the accommodation portion.
[0299] The foregoing description of the exemplary embodiments of the present invention has been provided for the purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise forms disclosed. Obviously, many modifications and variations will be apparent to practitioners skilled in the art. The embodiments were chosen and described in order to best explain the principles of the invention and its practical applications, thereby enabling others skilled in the art to understand the invention for various embodiments and with the various modifications as are suited to the particular use contemplated. It is intended that the scope of the invention be defined by the following claims and their equivalents.

Claims

1. A developer container comprising: an accommodation portion that accommodates a developer; a discharge port through which the developer is discharged from an outer peripheral portion of the accommodation portion on a front end side in a mounting direction; an opening and closing portion that opens and closes the discharge port from an outside; and an attachment portion that surrounds the discharge port and to which the opening and closing portion is movably attached, wherein the attachment portion has an extending portion extending toward the front end side in the mounting direction, and the extending portion is provided with a support portion that supports the extending portion on a surface opposite to the opening and closing portion.
2. The developer container according to claim 1, wherein the accommodation portion includes an opposing structural portion having a portion facing the extending portion at an inner part on the front end side in the mounting direction, and the support portion is provided in a form connected to the opposing structural portion.
3. The developer container according to claim 1, wherein the support portion has a protruding portion that protrudes toward the front end side with respect to an end portion of the extending portion on the front end side in the mounting direction.
4. The developer container according to claim 1, wherein the support portion is provided with an identification shape portion for distinguishing whether or not the developer container can be mounted according to differences in types of the developer and the accommodation portion.
5. The developer container according to claim 2, wherein the support portion has a plate shape, and the plate-shaped support portion is configured as a portion that is fitted into a recess portion of a mounting destination when the accommodation portion is mounted.
6. The developer container according to claim 2, wherein the support portion has a plate shape, and the facing portion of the opposing structural portion has a curved surface shape of which a projection side is connected to the plate-shaped support portion or has a two-surface shape that is developed to cross diagonally.
7. The developer container according to claim 3, wherein the accommodation portion includes an opposing structural portion having a portion facing the extending portion at an inner part on the front end side in the mounting direction, and the protruding portion is provided in a form that is present on an upstream side in the mounting direction with respect to an end portion of the facing portion on a downstream side in the mounting direction.
8. The developer container according to claim 4, wherein the support portion is provided as a plurality of support portions at intervals, and the identification shape portion is provided in each of the plurality of support portions.
9. The developer container according to claim 1, wherein the accommodation portion is provided

with an additional identification shape portion for distinguishing whether or not the developer container can be mounted according to differences in types of the developer and the accommodation portion at an inner part on the front end side in the mounting direction.

10. An image forming apparatus comprising: an image forming section that forms an image consisting of a developer; and a developer container that accommodates the developer supplied to the image forming section; and a mounting portion in which the developer container is attachably and detachably mounted by inserting and removing the developer container, wherein the developer container is the developer container according to claim 1.

11. An image forming apparatus comprising: an image forming section that forms an image consisting of a developer; and a developer container that accommodates the developer supplied to the image forming section; and a mounting portion in which the developer container is attachably and detachably mounted by inserting and removing the developer container, wherein the developer container is the developer container according to claim 2.

12. An image forming apparatus comprising: an image forming section that forms an image consisting of a developer; and a developer container that accommodates the developer supplied to the image forming section; and a mounting portion in which the developer container is attachably and detachably mounted by inserting and removing the developer container, wherein the developer container is the developer container according to claim 3.

13. An image forming apparatus comprising: an image forming section that forms an image consisting of a developer; and a developer container that accommodates the developer supplied to the image forming section; and a mounting portion in which the developer container is attachably and detachably mounted by inserting and removing the developer container, wherein the developer container is the developer container according to claim 4.

14. An image forming apparatus comprising: an image forming section that forms an image consisting of a developer; and a developer container that accommodates the developer supplied to the image forming section; and a mounting portion in which the developer container is attachably and detachably mounted by inserting and removing the developer container, wherein the developer container is the developer container according to claim 5.

15. An image forming apparatus comprising: an image forming section that forms an image consisting of a developer; and a developer container that accommodates the developer supplied to the image forming section; and a mounting portion in which the developer container is attachably and detachably mounted by inserting and removing the developer container, wherein the developer container is the developer container according to claim 6.

16. An image forming apparatus comprising: an image forming section that forms an image consisting of a developer; and a developer container that accommodates the developer supplied to the image forming section; and a mounting portion in which the developer container is attachably and detachably mounted by inserting and removing the developer container, wherein the developer container is the developer container according to claim 7.

17. The image forming apparatus according to claim 10, wherein the mounting portion is provided with a recess portion into which a support portion is fitted.

18. The image forming apparatus according to claim 10, wherein, in a case where the developer container is the developer container according to claim 4, the mounting portion is provided with an identification matching portion that has a shape that matches the identification shape portion and enables mounting of the accommodation portion.

19. The image forming apparatus according to claim 10, wherein, in a case where the developer container is the developer container according to claim 8, the mounting portion is provided with a plurality of identification matching portions that have shapes that respectively match the identification shape portions of the plurality of support portions and enable mounting of the accommodation portion.

20. The image forming apparatus according to claim 10, wherein, in a case where the developer container is the developer container according to claim 9, the mounting portion is provided with an additional identification matching portion that has a shape that matches the additional identification shape portion and enables mounting of the accommodation portion.